

Theory of Heart: Decomposing Theory of Mind with Mechanistic Probes

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Abstract

Recent work shows activation steering substantially improves language models’ Theory of Mind (ToM) (Bortoletto et al. 2024), yet *what* changes remains unclear. We decompose ToM by comparing steered versus baseline activations using linear probes trained on 45 cognitive actions. Applying Contrastive Activation Addition (CAA) to 1,000 BigToM false-belief scenarios, we find successful ToM enhancement (32.5% to 46.7% accuracy) is mediated by activations processing emotional content—*emotion_perception* (+2.23), *emotion_valuing* (+2.20), *noticing* (+2.08)—while suppressing analytical processes—*questioning* (-0.78), *convergent_thinking* (-1.59). This reveals a “Theory of Heart”: successful perspective-taking is mediated by emotional understanding, not analytical reasoning.

Introduction

Theory of Mind (ToM)—the capacity to attribute mental states to oneself and others—represents a critical capability for AI systems engaged in social reasoning, alignment, and human collaboration. Recent benchmarking work by Bortoletto et al. (2024) has demonstrated that activation steering techniques, particularly Contrastive Activation Addition (CAA), can substantially improve language models’ ToM performance, often achieving accuracy improvements exceeding 10 percentage points. This raises a fundamental question: *what cognitive processes change* when models successfully engage in perspective-taking?

Traditional evaluations treat ToM as a monolithic capability measured through binary accuracy on belief attribution tasks. However, cognitive science views ToM as comprising multiple interacting processes: perceptual encoding, emotional inference, hypothesis generation, and counterfactual reasoning. If steering vectors modulate these components differently, comparing steered versus baseline activation patterns could reveal which processes are *essential* for successful ToM—effectively decomposing perspective-taking into its constituent building blocks.

We introduce a mechanistic decomposition approach combining linear probes trained on 45 cognitive actions with CAA steering vectors. By measuring which cognitive processes are systematically upregulated or downregu-

lated when ToM performance improves, we can identify the fundamental components underlying successful perspective-taking. This moves beyond “does steering work?” to “*why* does steering work, and what does this reveal about ToM itself?”

Contributions: (1) A mechanistic decomposition methodology using cognitive action probes to analyze steering effects; (2) Large-scale evaluation on 1,000 BigToM false-belief scenarios showing 14.2 percentage point accuracy improvement (32.5% to 46.7%); (3) Discovery that successful ToM systematically upregulates emotional/creative processes while downregulating analytical processes; (4) Evidence for a “Theory of Heart”—emotional understanding as the fundamental building block of perspective-taking in language models.

Methodology

Cognitive Action Taxonomy. We defined 45 cognitive actions across four categories: *Metacognitive*, *Analytical*, *Creative*, and *Emotional* (see appendix).

Probe Training Data Generation. We generated 31,500 synthetic examples (700 per action) using Gemma-3-4B—first-person narratives (2-4 sentences) varied across 20 everyday contexts, producing balanced datasets with realistic language patterns.

Activation Capture and Probing. Activations were extracted from layers 10-30 of Gemma-3-4B using nnsight (Alain and Bengio 2016). Inputs were augmented with suffix “The cognitive action being demonstrated here is” for consistent final-token extraction. We trained 45 binary linear probes using **one-vs-rest** with AdamW optimization, cosine annealing, and early stopping based on AUC-ROC.

CAA Steering Vector Training. We trained CAA steering vectors (Rimsky et al. 2023; Zou et al. 2023) on 752 contrastive triplets from BigToM’s *forward_belief* scenarios (50/50 false-true split). Each triplet contains story context, belief question, and positive (correct ToM) versus negative (incorrect ToM) completions, capturing representational differences between accurate and inaccurate perspective-taking. Vectors were trained across layers 14-30 using PCA-centered activation differences.

ToM Decomposition Methodology. We evaluated 1,000 false-belief scenarios from BigToM (*forward_belief_false*), comparing baseline versus steered conditions. Cognitive ac-

tion activations were captured at three timepoints: (1) at question, (2) after true answer, (3) after wrong answer. For each action, we computed layer count (layers 10-20 where probe confidence indicated presence) and analyzed baseline-steered differences to identify which cognitive processes mediate ToM improvements.

Results

Probe Performance. Binary probes achieved 0.78 average AUC-ROC and 0.68 F1 across 45 actions, confirming linear separability of cognitive representations (Alain and Bengio 2016). Mid-layer performance (layers 5-24) outperformed early/late layers, informing our layer 10-20 analysis window.

ToM Steering Efficacy. CAA steering on 1,000 false-belief scenarios improved accuracy from 32.5% to 46.7% (14.2pp gain), shifting 217 examples from incorrect to correct predictions.

Cognitive Decomposition of ToM. Successful ToM is mediated by robust upregulation of emotional processes: *emotion_perception* (mean $\Delta=+1.73$), *noticing* (+1.48), and *hypothesis_generation* (+1.63) remained elevated across all timepoints.

Conversely, analytical processes decreased: *questioning* ($\Delta=-1.24$), *convergent_thinking* (-0.00 to -1.80), and *understanding* (-0.77). This reveals successful ToM suppresses deliberate analytical interrogation while amplifying emotional inference.

Category-Level Analysis. Category aggregation shows emotional (+0.35, +0.20, +0.22) and creative processes (+0.35, +0.28, +0.24) consistently increase across timepoints, while analytical processes decrease (+0.06, -0.19, -0.19), confirming ToM is mediated by emotional grounding rather than analytical reasoning.

Discussion

Theory of Heart: Emotional Foundations of Perspective-Taking. Our decomposition reveals successful ToM is mediated by emotional understanding rather than analytical reasoning—a “Theory of Heart” where emotional grounding constitutes perspective-taking’s fundamental building block.

Implications for AI Cognition. This challenges assumptions that LLM social reasoning relies on deliberate chain-of-thought. Instead, ToM operates through rapid emotional inference—mirroring “hot” (affective) versus “cold” (deliberative) cognition. Decreased *questioning* and *convergent_thinking* suggest explicit interrogation may interfere with perspective-taking.

Building on Activation Steering. We extend Bortoletto et al. (2024) by revealing *why* steering works: amplifying emotional-perceptual substrate while suppressing analytical interference. This enables more targeted interventions emphasizing emotional inference.

Limitations. Synthetic probe data, single model (Gemma-3-4B), single BigToM condition, independence assumptions between actions, and layer count metrics may not capture activation magnitude. Findings require validation across diverse ToM scenarios.

Broader Impact. Implications for AI safety: cognitive monitoring could detect inappropriate emotional inference. However, emotional understanding capabilities raise concerns about potential misuse in persuasion or deception.

Conclusion

We mechanistically decompose Theory of Mind by comparing steered versus baseline activations. Building on Bortoletto et al. (2024), we reveal *what* changes: successful perspective-taking is mediated by emotional understanding rather than analytical reasoning. This “Theory of Heart”—that emotional processes mediate ToM—challenges assumptions about deliberative social cognition and suggests cognitive action probes offer a principled approach for decomposing complex AI capabilities.

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Appendix

Complete Cognitive Action Taxonomy

The 45 cognitive actions used for probe training were derived from established cognitive psychology and emotion research frameworks, then organized into five functional categories. Actions were systematically sampled from Bloom's Taxonomy (Anderson et al. 2001), Guilford's Structure of Intellect (Guilford 1967), Metacognitive Processes Framework (Flavell 1979), Krathwohl's Affective Domain (Krathwohl, Bloom, and Masia 1964), Gross's Process Model of Emotion Regulation (Gross 1998), and the Mayer-Salovey Emotional Intelligence Model (Mayer and Salovey 1997).

Metacognitive (7 actions):

- *reconsidering*: reconsidering a belief or decision
- *updating_beliefs*: updating mental models or beliefs
- *suspending_judgment*: suspending judgment and staying with uncertainty
- *meta_awareness*: reflecting on one's own thinking process
- *metacognitive_monitoring*: tracking one's own comprehension
- *metacognitive_regulation*: adjusting thinking strategies
- *self_questioning*: interrogating one's own understanding

Analytical (16 actions):

- *noticing*: noticing a pattern, feeling, or dynamic
- *pattern_recognition*: recognizing recurring patterns across situations
- *zooming_out*: zooming out for broader context
- *zooming_in*: zooming in on specific details
- *questioning*: questioning an assumption or belief
- *abstracting*: abstracting from specifics to general patterns
- *concretizing*: making abstract concepts concrete and specific
- *connecting*: connecting disparate ideas or experiences
- *distinguishing*: distinguishing between previously conflated concepts
- *perspective_taking*: taking another's perspective or temporal view
- *convergent_thinking*: finding the single best solution
- *understanding*: interpreting and explaining meaning
- *applying*: using knowledge in new situations
- *analyzing*: breaking down into components
- *evaluating*: making judgments about value or effectiveness
- *cognition_awareness*: becoming aware and comprehending

Creative (6 actions):

- *creating*: generating new ideas or solutions
- *divergent_thinking*: generating multiple creative solutions
- *hypothesis_generation*: generating possible explanations

- *counterfactual_reasoning*: engaging in 'what if' thinking
- *analogical_thinking*: drawing analogies between domains
- *reframing*: reframing a situation or perspective

Emotional (15 actions):

- *emotional_reappraisal*: reinterpreting emotional meaning
- *emotion_receiving*: becoming aware of emotions
- *emotion_responding*: actively engaging with emotions
- *emotion_valuing*: attaching worth to emotional experiences
- *emotion_organizing*: integrating conflicting emotions
- *emotion_characterizing*: aligning emotions with core values
- *situation_selection*: choosing emotional contexts deliberately
- *situation_modification*: changing circumstances to regulate emotion
- *attentional_deployment*: directing attention for emotional regulation
- *response_modulation*: modifying emotional expression
- *emotion_perception*: identifying emotions in self/others
- *emotion_facilitation*: using emotions to enhance thinking
- *emotion_understanding*: comprehending emotional complexity
- *emotion_management*: regulating emotions in self/others
- *accepting*: accepting and letting go of control

Memory (1 action):

- *remembering*: recalling relevant information or experiences

CAA Training Data Example

The CAA steering vectors were trained on contrastive pairs distinguishing correct from incorrect belief attribution. Each training triplet consists of a story with observational information, a question about the protagonist's belief, and two completions representing accurate vs. inaccurate Theory of Mind reasoning.

False Belief Example:

Story: "Noor is working as a barista at a busy coffee shop. Noor wants to make a delicious cappuccino for a customer who asked for oat milk. Noor grabs a milk pitcher and fills it with oat milk. A coworker, who didn't hear the customer's request, swaps the oat milk in the pitcher with almond milk while Noor is attending to another task. Noor does not see her coworker swapping the milk."

Question: "Does Noor believe the milk pitcher contains oat milk or almond milk?"

Positive (correct ToM): "Noor believes the milk pitcher contains oat milk."

Negative (incorrect ToM): "Noor believes the milk pitcher contains almond milk."

The key distinction is that the positive completion always reflects accurate perspective-taking (understanding what the

protagonist actually believes based on their observational access), while the negative completion represents a ToM failure (confusing the protagonist's belief with objective reality or making incorrect inferences).

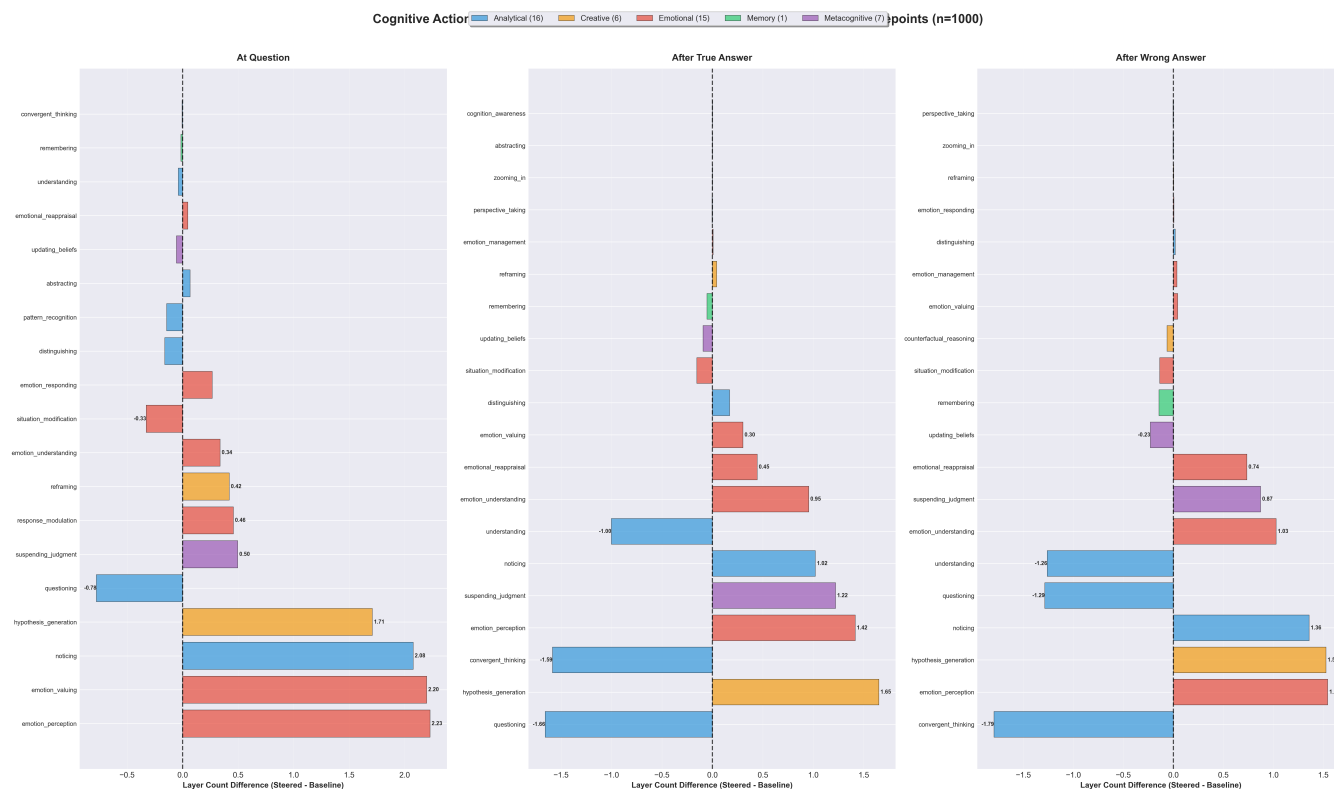


Figure 1: Steering effects across all cognitive actions and timepoints (n=1000). Left panels show individual action changes at three timepoints: at question (before answer), after true answer, and after wrong answer. Bars indicate mean layer count difference (steered - baseline) with positive values (right) showing increases and negative values (left) showing decreases. Emotional actions (*emotion_perception*, *emotion_valuing*, *noticing*) consistently increase across timepoints, while analytical actions (*questioning*, *convergent_thinking*, *understanding*) consistently decrease, revealing the emotional foundation of successful ToM.

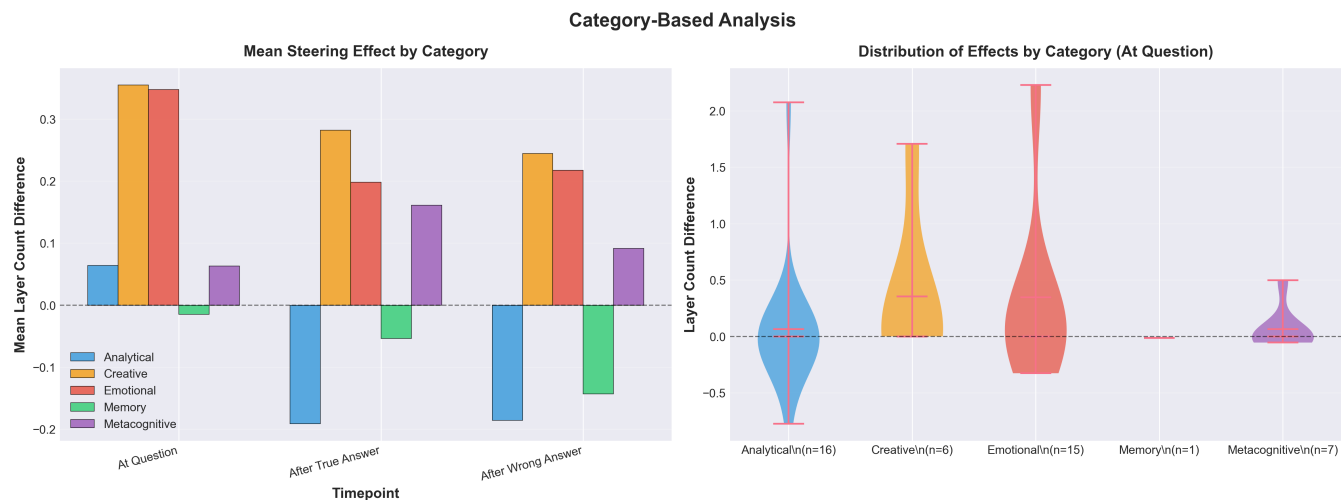


Figure 2: Category-level analysis of steering effects. Top left: mean steering effect by cognitive action category shows emotional and creative categories increase while analytical decreases. Top right: distribution of effects at question timepoint with emotional and creative categories shifted positive. Bottom left: proportion of 45 actions in each category. Bottom right: category $\times$ timepoint heatmap showing consistent patterns—emotional/creative increases persist across all timepoints while analytical decreases after answers. This systematic category-level pattern supports the "Theory of Heart" interpretation.

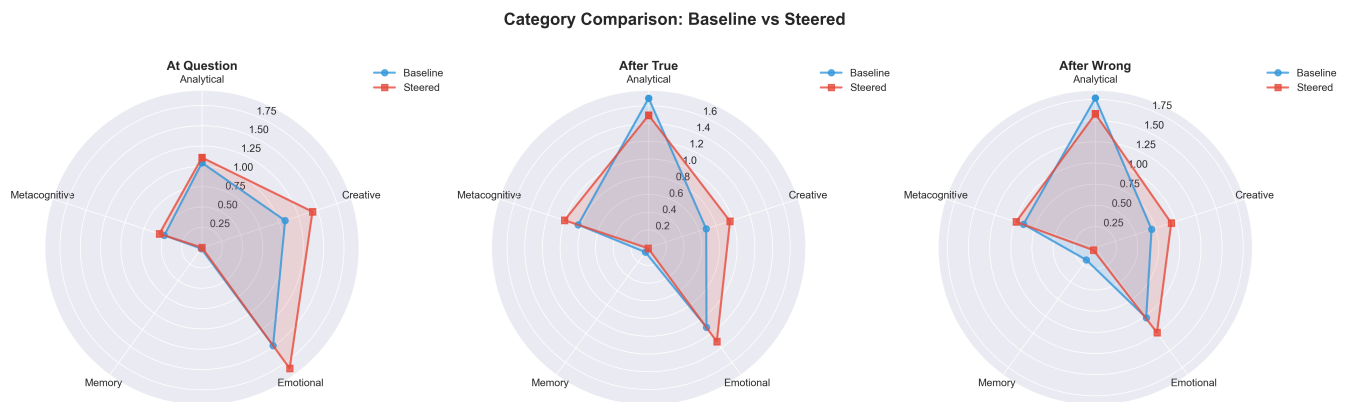


Figure 3: Radar chart comparing baseline versus steered cognitive action activation patterns across categories. The steered condition (red) shows elevated emotional and creative process engagement compared to baseline (blue), while analytical processes remain similar or decrease. This visualization highlights the shift toward emotional understanding that characterizes successful Theory of Mind enhancement through CAA steering.

Comprehensive Heatmap: Active Cognitive Actions × Timepoints\nGrouped by Category (n=1000 samples, 24 active actions)

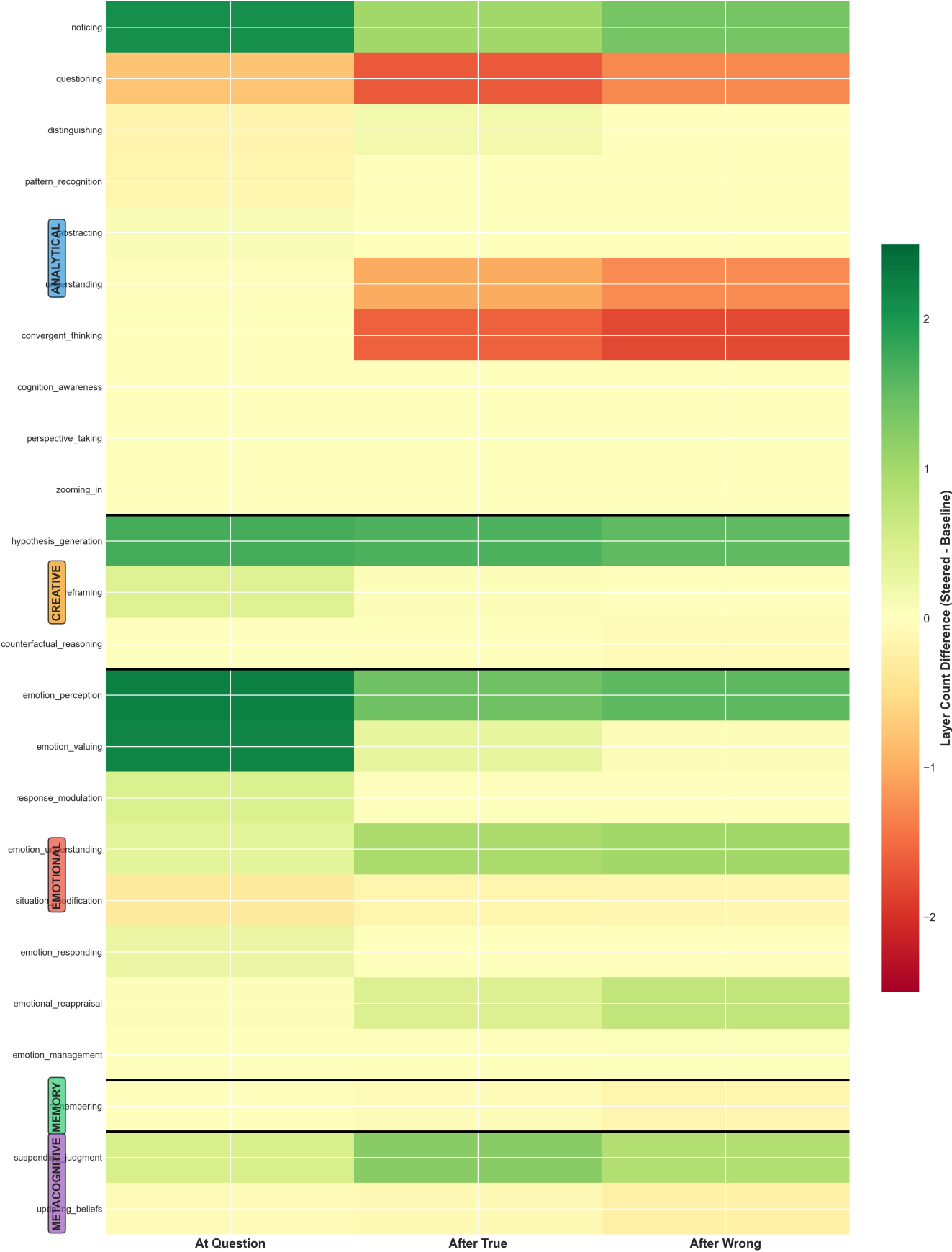


Figure 4: Heatmap of cognitive action activation differences (steered - baseline) across timepoints. Each row represents a cognitive action, and columns show the three measurement timepoints: at question, after true answer, and after wrong answer. Red indicates upregulation and blue indicates downregulation. The consistent patterns across timepoints reveal that emotional actions (top section) are systematically upregulated while analytical actions show more variable and often negative modulation, supporting the emotional foundation of ToM.

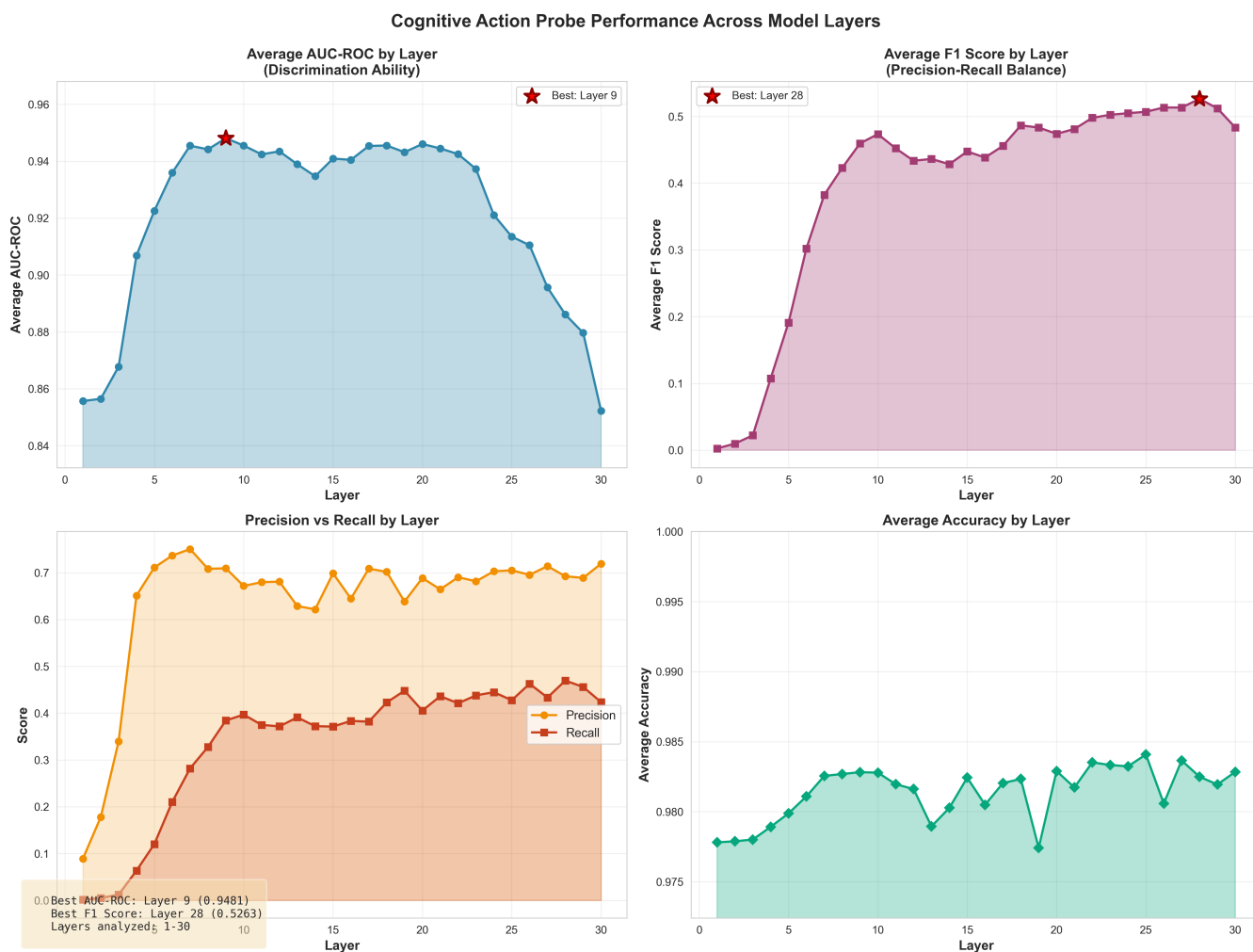


Figure 5: Cognitive action probe performance across all 30 layers of Gemma-3-4B. The visualization shows average AUC-ROC scores for each layer, with Layer 9 achieving peak performance (0.948 AUC-ROC). Strong performance is maintained across mid-layers (5-24), while early and late layers show degraded performance. This pattern suggests that early layers focus on surface-level features, mid-layers capture high-level cognitive abstractions, and late layers optimize for next-token prediction, potentially overwriting intermediate representations.

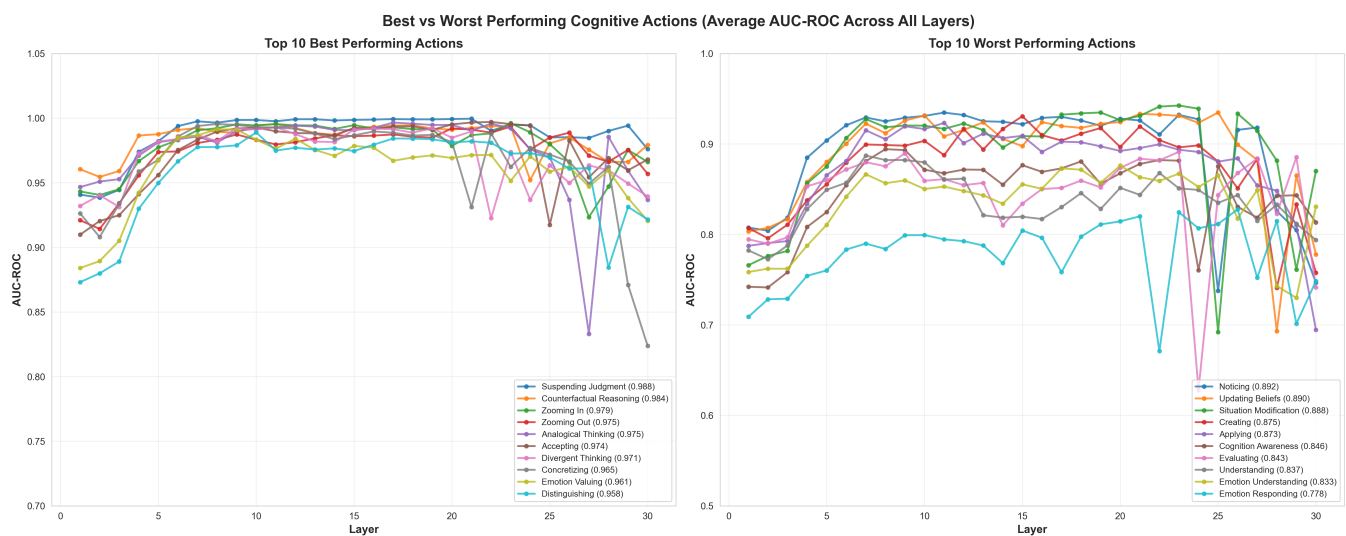


Figure 6: Comparison of top 10 and bottom 10 performing cognitive actions ranked by average AUC-ROC across all layers. Best performers like *suspending_judgment* (0.988) and *counterfactual_reasoning* (0.984) show consistently high performance and distinct activation patterns across most layers. Worst performers like *emotion_responding* (0.778) and *understanding* (0.837) exhibit more variability and lower overall discrimination ability, suggesting these concepts may be more distributed or context-dependent in the model's representation space.